Ayush Suryawanshi

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PROFESSIONAL SUMMARY

Chemical engineer with an M.Tech in Catalysis Technology from IIT Madras and a year of hands-on electrocatalyst work. Built and tested Bi-modified Pt/C anode catalysts for formic acid fuel cells, from synthesis and ink formulation to electrode preparation and potentiostat testing. Spent a semester at NTNU, Norway, where the synthesis had to be reworked after leftover NaBH4 ignited during filtration. Currently a Management Trainee at LOXIM Industries, evaluating electrocoagulation for effluent pretreatment. Keen to work on AEM electrolyzer electrodes.

EDUCATION

Master of Technology, Catalysis Technology (Chemical Engineering) — CGPA 8.51/10 | Jun 2024 – May 2026
Indian Institute of Technology Madras, Chennai

Bachelor of Technology, Chemical Engineering — CGPA 8.17/10, First Division with Honours | 2019 – 2023
Indore Institute of Science & Technology, Indore, affiliated to RGPV Bhopal

Senior School Certificate, Class XII, CBSE — 84.2% | 2019
Delhi Public School, Chhindwara

PROFESSIONAL EXPERIENCE

Management Trainee | Jul 2026 – Present
LOXIM Industries Ltd., Ahmedabad

- Evaluated electrocoagulation and electroflocculation as pretreatment options in a sulfate removal project.
- Leading technology selection for sulfate removal from plant effluent. Options shortlisted so far point to a projected **50–60%** lower OPEX, from INR 2.5/L to about INR 1.0–1.2/L.
- Running screening, sample trials and feasibility checks with more than ten technology vendors.
- Trained on HPLC, GC-MS, FTIR and UV-Vis; completed plant safety induction and HSE training.

RESEARCH PROJECTS

Bi-Modified Pt/C Catalysts and Quinone Mediation for Formic Acid Fuel Cells | May 2025 – May 2026
M.Tech Thesis | Guide: Prof. Raghuram Chetty, IIT Madras

- Made four Bi-modified Pt/C catalysts at Bi precursor levels from 1.25 to 10 mM, using aqueous adsorption, NaBH4 reduction and a 300 °C anneal under argon.
- Cutting the Nafion in the catalyst ink from 20 µL to 5 µL gave **2.4–2.8×** higher peak current; the thicker ionomer film had been slowing formic acid diffusion to the catalyst.
- Prepared electrodes on glassy carbon and carbon paper and tested them in a three-electrode cell on an Autolab potentiostat, measuring CV response, ECSA and mass activity.
- Bi(1.25)-Pt/C kept **92.9%** of its current over a one-hour hold at 0.55 V vs RHE, against 58.5% for commercial Pt/C. With 1 mM AQSA mediator the figures were 70.4% and 55.6%.
- Used FE-SEM, EDX, XRD and BET to connect Bi loading with particle morphology, lattice strain and activity.

Catalyst Synthesis Rework at NTNU | Aug 2025 – Dec 2025
Research stay, NTNU, Norway, on a UTFORSK scholarship | Host: Prof. Edd Anders Blekkan

- The first batches lost material during filtration, and in the second batch leftover NaBH4 ignited on the filter.
- Moved to an aqueous NaBH4 route with less reductant and a changed filtration step. The ignition risk went away and the third batch gave the highest yield, about 220 mg.

DFT Study of the Oxygen Reduction Reaction on a Pt13 Cluster | Jan 2025 – May 2025
Course Project, Computational Methods in Catalysis | Guide: Dr. Jithin John Varghese, IIT Madras

- Optimized O2, OOH*, O*, OH* and H2O on a Pt13 cluster in Gaussian 16; every step of the four-electron pathway came out downhill in SCF energy.

SNCR Retrofit for NOx Reduction in a Biomass Boiler | Aug 2025 – Nov 2025
Process Design Project, NTNU | Industry partners: Billerud Viken and Vardar Varme

- Co-authored a four-person design study. Combustion and NOx kinetics were modelled in Python with Cantera and pointed to about 60% NOx reduction.

TECHNICAL SKILLS

Electrochemical Testing: Autolab potentiostat, three-electrode cell, CV, LSV, EIS, chronoamperometry, chronopotentiometry, ECSA, durability testing

Electrodes and Catalysts: Wet-chemical catalyst synthesis, catalyst ink formulation, ionomer loading, drop-cast electrodes on glassy carbon and carbon paper

Materials Characterization: SEM-EDX, XRD, BET, XPS, TGA, FTIR, UV-Vis

Software: OriginPro, Excel, Python, Gaussian 16, Avogadro

Lab Safety: Handling acids and reducing agents, inert-gas work, plant HSE and SOP training

Relevant Coursework: Catalyst Preparation and Characterization Lab, Experimental Methods in Catalysis, Transport Phenomena, Process Safety and Hazards Management

POSITIONS OF RESPONSIBILITY

Deputy Placement & Internship Coordinator | Aug 2024 – Jul 2025

Placement & Internship Cell, IIT Madras

- Handled placements and internships for the 2023–25 and 2024–26 M.Tech batches; 75% of M.Tech chemical engineers were placed.

CONFERENCES & WORKSHOPS

- Presented a poster on SNCR NOx reduction at Treforedlingsforum 2025, Oslo, November 2025.
- GIAN course on characterization of heterogeneous catalysts, IIT Madras, April–May 2025.
- Paper presentation at SCHEMCON 2021, MANIT and IISER Bhopal, October 2021.

AWARDS & SCHOLARSHIPS

- GATE 2024, Chemical Engineering: All India Rank 864 of 13,937 candidates.
- UTFORSK mobility scholarship from NTNU for a research semester in Norway, August–December 2025.